NAVAL ARCHITECTURE
FREE SURFACE EFFECT

1 A ship of 6000 tonne displacement has its centre of gravity 5.9 m above the keel and transverse metacentre 6.8 m above the keel. A rectangular double bottom tank which is 10.5 m long, 12 m wide and 1.2 m deep is now half-filled with sea water of density 1025 kg/m3. Calculate the metacentric height in this condition.

[0.724m]

2 A double bottom tank is 23 m long. The half breadths of the top of the tank are 5.5, 4.6, 4.3, 3.7 and 3.0 m respectively. When the ship displaces 5350 tonne, the loss in metacentric height due to free surface is 0.2 m. Calculate the density of the liquid in the tank.

[0.8867 tonne/m^3]

3 A tanker 25 000 tonnes displacement and 24m wide is carrying oil of relative density 0.8 in nine cargo tanks each of 10m length. Calculate the free surface effect in the case when:

a. The tanks have no longitudinal bulkheads.
b. Each tank has one central longitudinal bulkhead.
c. Each tank is subdivided into three equal parts by two longitudinal bulkheads.
d. Each tank is subdivided into three parts by two longitudinal bulkheads with the centre tank having twice the width of the wing tanks.

[3.318m; 0.829m; 0.369m; 0.518m]